

**STRULE AUTOMATION**

Industrial Controls · Bay Area &amp; Northern California

## TRAINING PRIMER

# VFD Fundamentals: How a Drive Runs a Motor

Walk any modern aggregate, materials recovery, or organics plant and you'll find gray boxes bolted next to motor control centers, humming, with a little display and a few buttons. Those are variable frequency drives (VFDs), and they're running a big share of your pumps, fans, blowers, and conveyors. Most techs learn them by trial and error: something trips, somebody power-cycles it, and everyone moves on. That works right up until it doesn't. This primer is the fundamentals, written the way I'd explain it standing at the panel with you, so the drive stops being a mystery box and starts being a tool you understand.

## Why the plant put a drive there in the first place

Start with the problem a VFD solves. An alternating current (AC) induction motor — the workhorse motor in your plant — wants to spin at one speed. Feed it the 60 hertz (Hz) power off the grid and it runs at roughly its nameplate speed, full stop, every time. Hertz just means cycles per second; it's how fast the AC power alternates back and forth. Sixty Hz gives you full speed. That's fine if the load always needs full speed. But a lot of loads don't.

Take a blower feeding air into a digester or a baghouse fan pulling dust. The process needs different airflow at different times. Without a drive, your only tricks are throttling a damper or a valve — which is like driving with your foot on the gas and the brake at the same time — or starting and stopping the motor. Both waste energy and beat up the equipment.

A VFD gives you three real wins:

- **Energy savings on pumps, fans, and blowers.** This is the big one, and I'll show you the math later. Slowing a fan down a little saves a lot of power.
- **Soft starting.** Throw a big motor straight across the line and it slams to full speed, pulling six to eight times its normal current for a moment and jerking the whole drivetrain. A drive ramps it up smooth, which cuts the electrical shock to your power system and the mechanical shock to belts, couplings, and bearings.
- **Process speed control.** You can dial the motor to exactly the speed the process wants, and change it on the fly from the control system.

## The one relationship to understand: volts per hertz

Here's the heart of it. To change a motor's speed, the drive changes the frequency it feeds the motor. Lower frequency, slower motor. Simple enough.

But frequency alone isn't the whole story. If you drop the frequency and leave the voltage high, you shove too much magnetism into the motor's iron and it overheats — the core saturates, the magnetizing current climbs, and the motor cooks. So the drive lowers the voltage right along with the frequency, keeping the ratio between them roughly constant. That ratio is called volts per hertz (V/Hz).

Think of it this way: voltage is the push and frequency is the pace. To keep the motor's torque — its twisting strength — steady, you have to keep the push in proportion to the pace. On a 460-volt, 60 Hz motor that's about 7.7 volts for every hertz. At 30 Hz the drive feeds roughly 230 volts, at 45 Hz roughly 345 volts, and so on. Keep that ratio and the motor makes about the same torque at any speed.

This holds up to **base speed** — the motor's nameplate frequency, usually 60 Hz. Below base speed, V/Hz stays constant and torque stays available. Above base speed you can raise frequency, but the drive can't push voltage past what the incoming line gives it, so the ratio falls and torque drops off. That's why running a motor faster than nameplate is a "check the spec and the load carefully" move, not a free lunch.

## What's actually inside the box: three stages

A VFD does its job in three stages. Once you can picture them, everything the drive does — and most of the ways it fails — makes sense.

**Stage one: the rectifier.** The incoming AC power hits a bank of diodes (one-way electrical valves) that flips the alternating current into direct current (DC). DC only flows one direction, like a battery. This stage is called the rectifier because it "rectifies" the wobbly AC into rough DC.

**Stage two: the DC bus.** That rough DC feeds into a set of large capacitors — components that store electrical energy, like small fast-charging batteries — that smooth it into a steady, high-voltage DC supply. This storage rail is the DC bus. On a 460-volt drive the bus sits around 650 to 680 volts DC. Hold that number in your head, because it's the one that can hurt you, and we'll come back to it under safety.

**Stage three: the inverter.** This is where the magic happens. The inverter takes that steady DC and chops it back into AC — but AC at whatever frequency and voltage the drive chooses. It does this with fast electronic switches called insulated-gate bipolar transistors (IGBTs). An IGBT is a transistor, meaning a solid-state switch with no moving parts, built to handle high power and switch on and off thousands of times a second. By flicking the DC on and off in a precise pattern, the IGBTs synthesize an AC waveform at the exact frequency the motor needs. Slow the switching pattern and you get lower frequency and a slower motor.

So the whole chain is: AC in, rectify to DC, store and smooth it on the bus, then invert it back to AC at the frequency you want. Grid speed goes in, any speed you like comes out.

## | The parameters a tech actually meets

You don't program a drive from scratch, but you will meet a handful of settings, and getting them wrong causes real grief. Here are the ones that matter.

**Motor nameplate values.** Before anything else, the drive needs to know the motor it's running: rated voltage, full-load amps (FLA), rated frequency, rated speed in RPM, and horsepower or kilowatts. These come straight off the motor's nameplate. Enter them wrong and everything downstream — the V/Hz curve, the overload protection — is working off bad information. Fat-finger the FLA and the drive either lets the motor overheat or trips for no reason.

**Accel and decel time.** These set how many seconds the drive takes to ramp from stop up to speed (accel) and back down to stop (decel). Set accel too short and the drive strains to move the load fast, drawing heavy current and tripping on overcurrent. Set decel too short and a spinning load acts like a generator, dumping energy back onto the DC bus and tripping on overvoltage. Longer ramps are gentler; you tune them to what the load can take.

**Minimum and maximum frequency.** These are the speed limits — the floor and ceiling the drive will run between. Set the minimum too low and some motors don't cool well at a crawl (many are fan-cooled by their own shaft) or a pump can't make enough head to move anything. Set the maximum above nameplate and you're into that reduced-torque, over-speed territory that can overload the motor or the driven equipment.

**Overload setting.** This is the drive's electronic protection for the motor, set from that full-load amps figure. It decides how much current, for how long, before the drive trips to save the motor from burning up. This is the one people are tempted to abuse, and I'll be blunt about it below.

## | A worked example: slowing a blower from 60 to 45 Hz

Say you've got a blower running at 60 Hz and the process only needs about three-quarters of the airflow. You bring the drive down to 45 Hz. What actually happens?

**Speed** tracks frequency almost directly. Forty-five is three-quarters of sixty, so the motor and blower slow to about three-quarters speed. Airflow from a blower or fan follows speed pretty closely, so you get roughly three-quarters of the air. That part's intuitive.

Here's the part that surprises people. For a fan, blower, or centrifugal pump, the power it takes to run drops with the **cube** of the speed. Cube means speed multiplied by itself three times. Three-quarters speed cubed is  $0.75 \times 0.75 \times 0.75$ , which is about 0.42. So dropping to three-quarters speed drops the power draw to roughly 42 percent — you're using less than half the energy to move three-quarters of the air.

That's the whole energy argument in one example. A modest slowdown you'd barely notice on the process saves you better than half the power. Trim a fan back 20 percent and you can cut its energy roughly in half. That relationship — the cube law — is why the drive paid for itself, and it only applies to fans, blowers, and centrifugal pumps. A conveyor or a positive-displacement pump is a different animal; slowing it saves energy roughly in proportion to speed, not to the cube. Know which kind of load you've got before you promise anyone savings.

## ■ The honest part: a VFD is not a mechanical fix

I'll say this plainly because I've watched it go wrong. A drive controls a motor. It does not fix what the motor is bolted to. If a bearing is failing, a belt is worn, a coupling is misaligned, an impeller is caked, or a screw conveyor is jammed with debris, the drive will show you the symptom — high current, nagging trips, a motor working harder than it should — but slowing it down or nudging a parameter only masks the problem while the real damage keeps going. When a drive starts complaining, the first question is always "what changed on the mechanical side?" before you ever touch a setting. More often than not the drive is just reporting a fault that started somewhere else on the machine.

## ■ Safety: the DC bus can kill you after the power is off

This is the one I need you to take dead seriously. Remember those big capacitors on the DC bus, sitting at 650-plus volts DC? They hold their charge after you kill power to the drive. Not for a second — for minutes. You can lock out the disconnect, walk up to a "dead" drive, open it, touch the bus, and be killed by energy that was stored inside before you ever threw the switch.

So the rule is simple and non-negotiable. Kill power and lock it out. Then **wait the manufacturer's rated discharge time** — printed on the drive, typically five minutes or more, so check the label on your specific unit. Then, before your hands go anywhere near the bus terminals, **verify zero volts** with a meter you trust, tested on a known live source first so you know the meter itself works. Wait, then verify. Never one without the other.

Any time you're taking a live reading on a drive or the power feeding it, you're doing energized work: right personal protective equipment (PPE) for the arc-flash rating of that gear, the right meter, and a second person aware of what you're doing. If a reading doesn't require the cover off and the power on, take the cover off with the power off. When it's time for hands-on mechanical work on the motor or driven equipment, that's full lockout/tagout (LOTO) and zero-energy verification, same as any other machine.

And one more, because it's a safety issue even though it looks like a settings issue: **do not make a nuisance trip go away by just widening the limit**. If the overload keeps tripping, the answer is not to raise the overload number until it stops. That protection is standing between a real fault and a burned-up motor or a fire. If you're genuinely convinced a setpoint is set tighter than the equipment's

true rating — say the FLA was entered low — the legitimate path is management of change: confirm the correct value against the nameplate, make one change, inside the rated limit, and log it. One change, documented, for a real reason. Never disable protection to silence an alarm.

## Try this on your own plant, safely

Next time you're near a drive that's running normally — no covers off, no tools, nothing energized opened — just read the display. Find the output frequency and see how far it is below 60 Hz. Find the output current and compare it to the motor's full-load amps on the nameplate. Get a feel for what "normal" looks like on your equipment when everything's healthy. That baseline is worth more than any troubleshooting trick, because the day something's wrong, you'll recognize it. Pair this primer with the companion piece, "VFD Keeps Tripping," and you'll have both the how-it-works and the what-to-do-when-it-doesn't.

## Glossary

- **Alternating current (AC):** Electrical power that reverses direction many times a second; standard grid and motor power.
- **Base speed:** The motor's nameplate frequency (usually 60 Hz), the top of the range where volts-per-hertz stays constant and full torque is available.
- **Capacitor:** A component that stores electrical energy and releases it quickly; the DC bus uses a bank of them.
- **DC bus:** The high-voltage direct-current storage rail inside the drive, roughly 650-680 volts on a 460-volt unit; holds a lethal charge after power-off.
- **Decel time:** Seconds the drive takes to ramp the motor from speed down to stop.
- **Accel time:** Seconds the drive takes to ramp the motor from stop up to speed.
- **Diode:** A one-way electrical valve; the rectifier is built from these.
- **Direct current (DC):** Electrical power that flows one direction only, like a battery.
- **Full-load amps (FLA):** The motor's rated running current at full load, taken from the nameplate; the basis for overload protection.
- **Hertz (Hz):** Cycles per second; the frequency of AC power, which sets motor speed.
- **IGBT (insulated-gate bipolar transistor):** A fast solid-state switch in the inverter that chops DC into variable-frequency AC.
- **Inverter:** The drive stage that converts DC back into AC at the chosen frequency and voltage.
- **Lockout/tagout (LOTO):** The procedure for isolating and verifying zero energy before hands-on work.
- **Overload setting:** The drive's electronic motor protection, set from full-load amps; trips on too much current for too long.

- **Personal protective equipment (PPE):** Gear rated for the hazard, including arc-flash-rated protection for energized work.
- **Rectifier:** The drive stage that converts incoming AC into rough DC.
- **Torque:** A motor's twisting strength.
- **Variable frequency drive (VFD):** The device that varies the frequency and voltage fed to an AC motor to control its speed.
- **Volts per hertz (V/Hz):** The ratio of voltage to frequency the drive holds roughly constant below base speed to keep torque available.

A closing note from the field: drives cause the most confusion where they get handed between vendors — one outfit set it up, another swapped the motor, a third rewrote the parameters — and nobody has the whole picture. That's the spot where a second set of eyes on the settings and the nameplate data earns its keep.